

Manjunath Popuri

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PROFESSIONAL SUMMARY

AI/ML Engineer with 4+ years of production experience building and scaling machine learning systems across agentic AI orchestration, LLM fine-tuning, RAG pipelines, and classical ML for financial risk and fraud detection. Proven track record owning the full model lifecycle - distributed training, feature engineering, inference optimization, serving infrastructure, and production monitoring - delivering measurable business impact at Stripe and enterprise scale at Cognizant across 70M+ daily transactions.

TECHNICAL SKILLS

Languages: Python, SQL, Java, Javascript, C, C++, R
AI/ML: PyTorch, TensorFlow, JAX, Hugging Face Transformers, XGBoost, Scikit-learn, LSTM, FinBERT, FinGPT (LLaMA-3), LLM Fine-tuning, QLoRA, PEFT, RLHF, SHAP, ONNX, INT8 Quantization, TensorRT, Embedding Models
Agentic AI & LLM Systems: LangChain, LangGraph, CrewAI, AutoGen, MCP, Tool Calling, Multi-Agent Orchestration, RAG Pipelines, Hybrid Retrieval, Vector Search, ReAct, Prompt Routing, LlamaIndex, Structured Output
MLOps: MLflow, Docker, Kubernetes (EKS), CI/CD (GitHub Actions), Airflow, Feature Stores (Feast), Model Registry, A/B Testing, Drift Monitoring, pytest, Canary Deployments, FastAPI, PySpark, Apache Kafka, Streamlit
Cloud & Databases: AWS (SageMaker, EC2, S3, Lambda, EKS, EMR), Azure ML, GCP (Vertex AI, BigQuery), PostgreSQL, MongoDB, ChromaDB, Snowflake, Pinecone, Redis, Prometheus, Grafana

PROFESSIONAL EXPERIENCE

AI/ML Engineer | Stripe, New York, NY Jul 2025 - Present

- Architected a **multi-agent LangGraph orchestration system** for real-time financial fraud triage across **70M+ daily transactions**, coordinating specialized retrieval, classification, and escalation agents; reduced **false-positive review rate by 22%** and cut analyst escalation time by 35%.
- Engineered a production **ML model pipeline** on **AWS SageMaker + EKS** with **ONNX-optimized inference** and **INT8 quantization**, achieving **sub-150ms p95 latency** for 10K+ concurrent requests - a **3x throughput improvement** over the baseline serving setup.
- Built a **Feast-compatible feature store integration** for real-time fraud signal aggregation, ensuring consistent feature computation across training and serving environments; eliminated **40% of train-serve skew incidents**.
- Deployed a **ReAct-based agentic AI system** with **Claude function calling** and Pydantic structured output for autonomous dispute classification, processing **10K+ cases/day** at **89% routing accuracy** and reducing manual review load by 28%.
- Scaled **high-availability serving infrastructure** on **AWS EKS** with **horizontal pod auto-scaling**, circuit-breaker patterns, and **multi-AZ failover**; sustained **99.9% SLA compliance** during 8x peak traffic surges with **zero-downtime rolling deployments**.

Graduate Research Assistant | Binghamton University, Binghamton, NY Aug 2024 - May 2025

- Fine-tuned **FinBERT** on **3,000+ Russell 3000 earnings call transcripts** using **GPU-accelerated training**, achieving a **14% F1 improvement** over the base model for risk-sentiment classification with full experiment reproducibility tracked in **MLflow**.
- Engineered an **end-to-end NLP pipeline** (tokenization, embedding generation, **ChromaDB vector indexing**, inference serving) reducing manual text-processing overhead by **25%** and enabling **semantic retrieval at sub-100ms latency** over **100K+ document chunks**.
- Designed a **LangGraph-based agentic summarization system** with **persistent memory** and **tool-calling** for real-time SEC filing retrieval, enabling multi-step reasoning over multi-quarter earnings sequences and reducing analyst review time by **30%**.
- Conducted responsible AI analysis using **SHAP attribution** and **counterfactual bias auditing** across sector and demographic slices; surfaced 3 systematic attribution errors and improved **model fairness scores by 12%** on held-out evaluation sets.

Machine Learning Engineer | Cognizant, Hyderabad, India Jul 2021 - Jun 2023

- Deployed **XGBoost + LSTM ensemble credit default models** on **10M+ customers** via **AWS SageMaker**; delivered an **8% AUC-ROC improvement** over the legacy rule-based system, measurably reducing credit loss provisions at enterprise scale.
- Architected **PySpark feature engineering pipelines** on **AWS EMR** with **Feast-compatible feature store** integration and automated validation; cut data processing time by **20%** and enabled near real-time risk scoring with consistent **train-serve feature parity**.
- Applied **ONNX model export** to production credit scoring endpoints, reducing **inference latency by 30%** on cost-optimized EC2 instances with no measurable accuracy degradation on holdout evaluation sets.
- Integrated production models into **AWS SageMaker CI/CD pipelines** with automated **data, prediction, and concept drift monitoring** plus **auto-retraining triggers**; maintained **99.8% system uptime** and zero undetected degradation events across 3 live models.

PROJECTS

Multi-Agent Recommendation & Personalization Platform | LangGraph, LLaMA-3, FastAPI, ChromaDB, Redis, Azure ML, MLflow

- Architected a **multi-agent LangGraph system** with specialized agents for user intent classification, real-time product retrieval, personalized ranking, and behavioral context management; integrated **Redis** for session memory and **Feast** for consistent user feature serving, improving simulated click-through rate by **34%** over a rule-based baseline.
- Deployed on **Azure Container Apps** with **MLflow model registry**, automated **canary deployments**, and **Prometheus/Grafana** observability; achieved **99.6% uptime** with automated rollback on ranking degradation and **A/B testing pipelines** validating recommendation quality across user cohorts.

Gen AI Earnings Call Analytics Engine | FinGPT (LLaMA-3), QLoRA, PEFT, RAG, LangChain, Streamlit, Docker, GCP (Vertex AI, Cloud Run), Power BI

- Fine-tuned **FinGPT (LLaMA-3)** with **QLoRA/PEFT** on 3,000+ earnings call transcripts for forward-looking risk summarization; achieved **91% accuracy** on held-out benchmarks and a **14% ROUGE-L improvement** over the zero-shot LLaMA-3 baseline.
- Built a production **RAG pipeline** with **LangChain + ChromaDB** using **hybrid dense + BM25 sparse retrieval**; reduced retrieval latency by **35%** vs. keyword search and improved semantic recall by **18% (MRR)**; deployed via **4-bit quantized FastAPI** serving on **GCP Cloud Run** with **sub-400ms median inference latency**.

Job Tracker - Chrome Extension | JavaScript, Google Drive API, Google Sheets API, Chrome MV3

- Built and published a **Chrome extension (Manifest V3)** that auto-extracts job details from any posting and logs them to a personal **Google Sheet and Drive folder** via **OAuth 2.0 (drive.file scope)**, keeping each user's data private in their own Google Drive.
- Shipped an **Applications dashboard** with search, sort, and real-time sync; published to the **Chrome Web Store** with custom onboarding flow, privacy policy, and MIT license.

EDUCATION

Binghamton University, State University of New York, Binghamton, NY
Master of Science - Computer Science (AI Specialization) | GPA: 3.54 / 4.0

Aug 2023 - May 2025